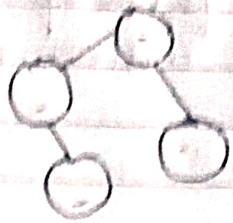


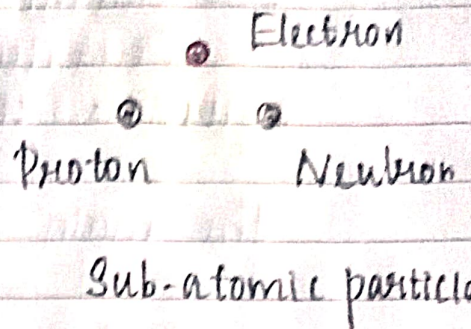
Quantum Physics: Quantum physics is a part of physics that describes the smallest things in our universe: molecules, atoms, sub atomic particles.



molecules



Atoms



Sub-atomic particle

Classical Physics: It works well for large objects, but at microscopic scales particles behave very differently.

Key Ideas in Quantum Physics:

- wave-particle duality: A particles can behave like both waves and particles.
- Superposition: a particle can exist in multiple states at the same time.
- Entanglement: two particles can remain connect even when far apart.
- Quantization: Energy exists in fixed packets called 'quanta'.
- Uncertainty principle: We cannot know certain properties (like position and momentum) exactly at the same time.

Example:

An electron does not orbit the nucleus like a planet around the Sun. Instead, it exists as a probability cloud.

//_

Quantum Mechanics: Quantum Mechanics is the mathematical theory used to describe and predict quantum physics phenomena.

- Quantum physics = the field of study
- Quantum Mechanics = the mathematical rules and equations used in that field.

The main contributors are,

- Max Planck
- Albert Einstein
- Neils Bohr
- Erwin Schrodinger
- Werner Heisenberg

The famous equation of Schrodinger wave equation:

$$i\hbar \frac{\partial}{\partial t} \psi = \hat{H} \psi$$

This equation predicts how quantum systems change over time.

Quantum Technology: Quantum Technology refers to technologies built using mechanics principles.

→ It uses the effects such as,

- superposition
- Entanglement
- Quantum tunneling.

→ It covers main area

- Quantum Computing
- Quantum Communication
- Quantum Cryptography
- Quantum Sensors
- Quantum Imaging

Example:

- Ultra-precise atomic clocks
- Quantum internet research
- Highly sensitive medical and navigation sensors

Quantum Application: Application of quantum science and technology includes

→ Medicine

- * Drug discovery
- * protein simulation
- * Faster medical data analysis.

→ Communication

- * Ultra-secure communication using quantum cryptography.

→ Banking and Security

- * Strong encryption system.
- * Fraud detection.

→ Artificial Intelligence

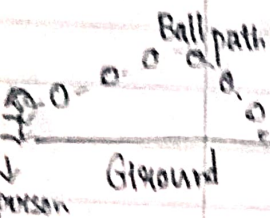
- * Faster optimization and machine learning.

→ Material Science:

- * Designing superconductor and new materials

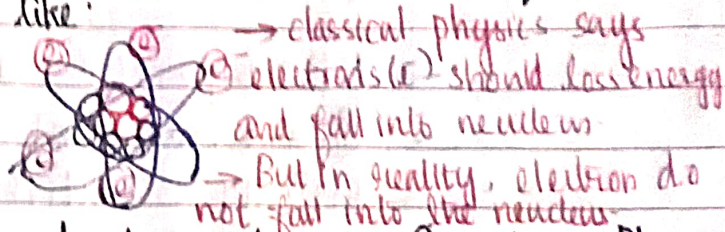
How classical physics Failed?

1. → Isaac Newton (1643/1687) - classical Physics, explained the motion of large object such as:
- cars
 - planets
 - cricket balls
 - Falling objects
- Developed laws of motion and gravity.
- Eg/ Throwing a ball its position, speed, landing point can be predicted accurately.



- It worked well for the macroscopic objects but failed to explain microscopic particles like:

- atoms structure
- electrons behavior
- light properties



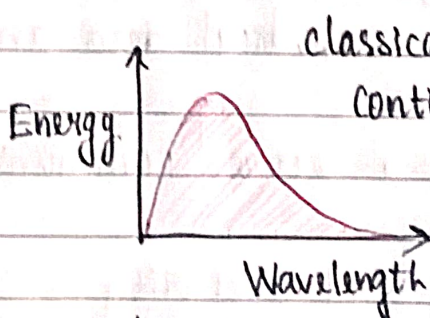
- This failure lead to the development of Quantum Physics

2. → Max Planck (1900) - Quantum theory: classical physics predicted infinite energy radiation from hot bodies (objects)

- Planck discovered Energy is emitted in small packets called quanta

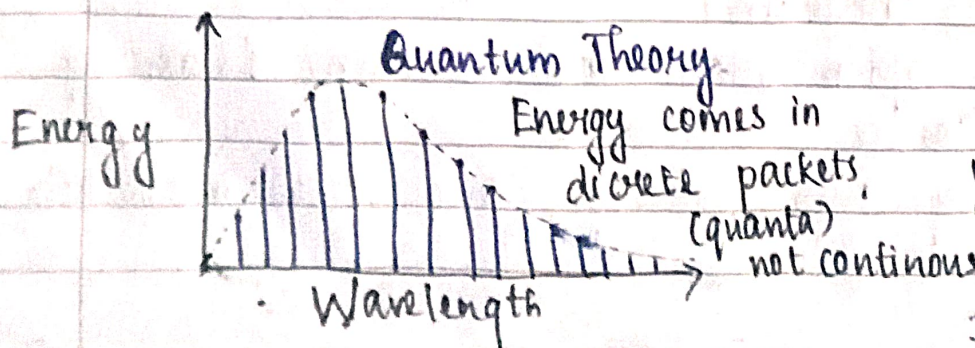


Hot object



classical prediction is continuous energy, which is infinite energy X

- Max Planck discovered that energy is emitted in small packets called quanta.



Quantum Theory

Energy comes in discrete packets (quanta) not continuous

Formula

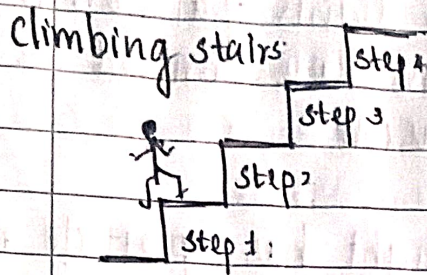
$$E = hf$$

where, E = energy of one quantum

h = Planck's constant ($6.626 \times 10^{-34} \text{ m}^2 \text{ kg/s}$)

f = frequency

→ Real-time Example:



⊗ Planck's idea of energy quanta solved the blackbody radiation problem and became the starting point of Quantum Physics.

- you move step by step, not continuously.
- Similarly, energy changes in fixed packets (quanta).

∴ Energy is not continuous, it is quantized and it comes in fixed packets called quanta.

3. → Albert Einstein (1905) - Photoelectric Effect [1879-1955]

- Classical wave theory could not explain electron emission from metals. This classical theory failed to explain:
 - * No electrons are emitted below a certain frequency, no matter how intense the light is.
 - * Electrons are emitted instantly, not with delay.

⊗ → Einstein proposed that light is made of tiny particles called photons. Photons have energy given

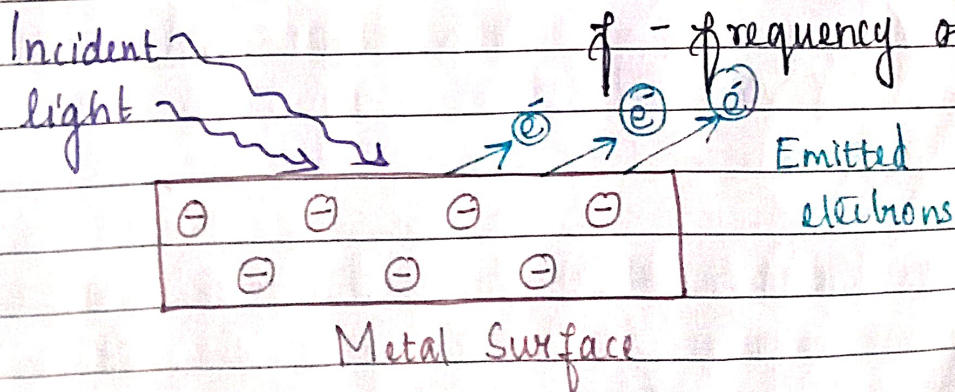
$$E = hf$$

Where

E = energy of photon

h = Planck's constant ($6.626 \times 10^{-34} \text{ J s}$)

f = frequency of light.



→ Real-Time Example

→ Solar calculator works, when light falls on it. Light shines on the solar cell → electrons are emitted → electric current is produced → calculator work.



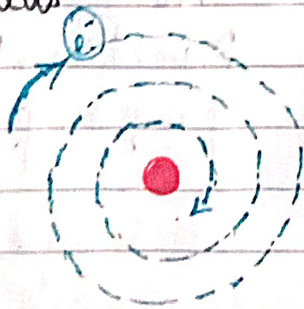
→ Einstein (1905) showed that light has particle nature (photons) through photoelectric effect.

→ Einstein's explanation proved that light behaves as particles. Einstein received that the Nobel Prize in Physics (1921) for this work. It laid the foundation for quantum physics.

→ The main key idea is $\times$ light is not only a wave, but also a particle. These particles are called photons. This dual nature of light is a special feature of quantum physics.

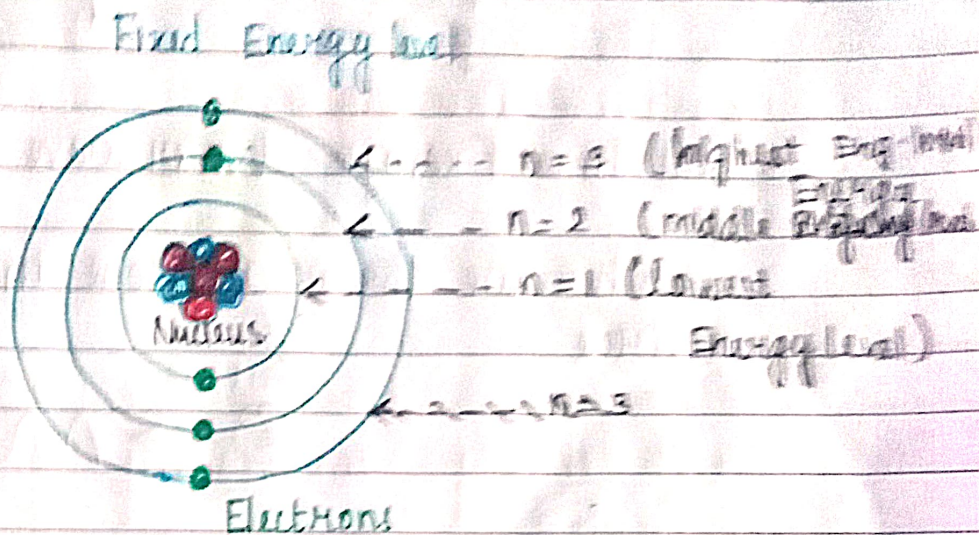
4. → Niels Bohr (1913) Bohr Atomic Model

→ According to classical physics, electrons should lose energy while orbiting the nucleus and finally falls into the nucleus.



Electrons should
fall into nucleus

→ Bohr proposed that electrons move around the nucleus in fixed circular paths called energy levels. Electrons do not lose energy in these orbits. Therefore, Bohr's atomic model says electrons move in fixed energy levels.



→ Real-Time Example:

- An elevator can stop only at the fixed floors (1, 2, 3, 4, 5). It cannot stop between floors.
- Similarly, electrons move only in fixed energy levels. They cannot exist between these levels.

→ Key Idea: Electrons occupy fixed energy levels around the nucleus and do not radiate energy while staying in these levels. So also explained the stability of atoms.

5. → LOUIS DE BROGLIE (1924) - Matter waves

→ He discovered particles can behave like waves. Louis de Broglie proposed that not only light, but also matters (particles) can behave like waves.



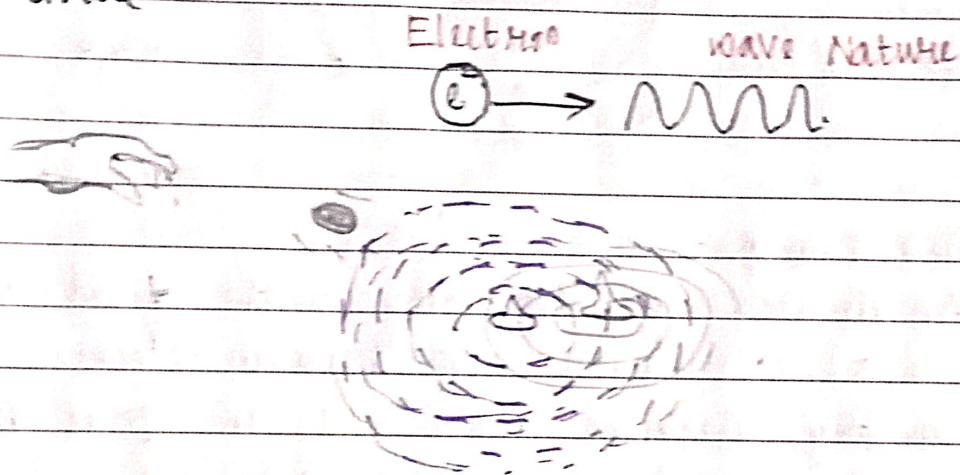
• This idea was confirmed by experiments with electrons

$$\lambda = \frac{h}{p}$$

Where λ = wavelength of matter (in metre)
 h = Planck's constant ($6.626 \times 10^{-34} \text{ J s}$)
 p = momentum of the particle.
 $p = mv$

→ Real-Time Example: Water wave Example

→ When we throw a stone in water, wave spread out in circle



→ Similarly, electrons and other tiny particles also show wave behavior.

→ This introduced the idea of wave-particle duality which is the foundation of modern quantum mechanics.

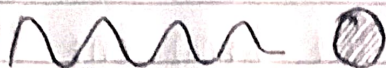
④ → Wave-Particle Duality: Both light and matter (like electrons) have dual nature - they behave as both particle and waves.

→ Louis de Broglie (1924) showed that matter can behave like waves, laying foundation of wave-particle duality in quantum physics.

→ Schrodinger (1926) gave a powerful equation that explains how quantum particles behave like waves, leading to the concept of superposition.

6 → ERWIN SCHRODINGER (1926) → Wave Equation

→ Erwin Schrodinger developed a mathematical equations to describe the behaviour of quantum particles as waves.



→ The time-dependent Schrodinger Equation is

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi$$

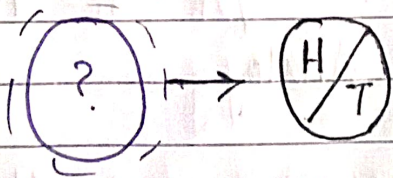
→ Schrodinger's equation tells us how the wave function ψ changes with time.

→ $|\psi|^2$ gives the probability of finding the particle at a certain position.

{ Where i = imaginary unit ($\sqrt{-1}$)
 $\hbar$ = Reduced Planck's constant ($h/2\pi$)
 ψ = Wave function
 $\hat{H}$ = Hamiltonian operator (total energy)
 t = time

→ Real-time Example (Spinning Coin Analogy)

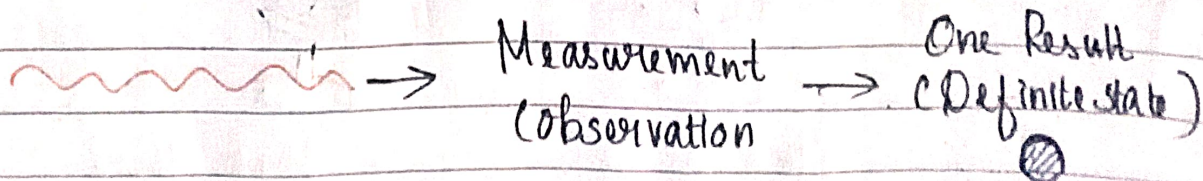
→ When a coin is spun, before it stops, it appears like both heads and tails at the same time.



Similarly, a quantum particle exists in a superposition of states before it is measured.

Key Idea

→ Superposition: A quantum system can be in multiple states at the same time until it is observed.

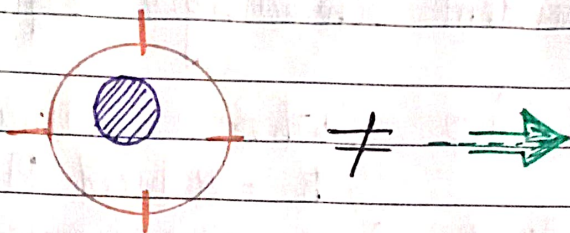


→ Nature has a limit, we cannot know everything at the same time.

4. → WERNER HEISENBERG (1927) → Uncertainty principle

→ Heisenberg (1927) discovered that we cannot know both the exact position and momentum (velocity $\times$ mass) of a particle at the same time.

→ The more accurate we know one, the less accurately we know the other.



→ Heisenberg uncertainty principle is expressed as

$$\Delta x \Delta p \geq \frac{h}{2}$$

Where, Δx = uncertainty in position

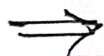
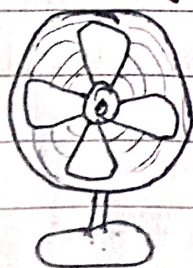
Δp = uncertainty in momentum

h = reduced plank's constant ($h/2\pi$)

$$h/2\pi = \hbar = 1.055 \times 10^{-34} \text{ J s}$$

Example: Spinning Fan Analogy

→ When a fan is spinning fast, its blades cannot be seen clearly.



Slow Speed
(Blades visible)

High Speed
(Blades blurred)

→ The faster it moves (momentum known), the less clearly we can see the blades (position unknown). We cannot know both exactly.

Working
Explanation

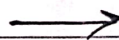
If we try to know the position (x) very exactly, then the momentum (p) becomes very uncertain. If we try to know the momentum very exactly, the position becomes uncertain. This is not the experimental error, but a fundamental property of nature.

- ✓ Shows the fundamental limit of measurement in the quantum world.
- ✓ Explains why electrons cannot have a fixed position and momentum together.
- ✓ Forms the basis of quantum mechanics.
- ✓ Used in many modern technologies like electron microscopes, semiconductors, quantum computers etc.

Quantum Particle



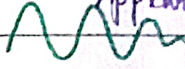
Many Possible States
(superposition)



Measurement
Attempt

or

uncertain
Appearance



→ Nature does not allow exact knowledge of both position and momentum at the same time.

Therefore

Heisenberg (1927) showed that nature has intrinsic limits on what we can measure, leading to the foundation of modern Quantum Physics.